Nabra at StanceEval-2026: Exploring Fine-Tuning Strategies for Arabic Stance Detection

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Abstract

We present Nabra, a system submitted to StanceEval-2026 Track 1 for Arabic stance detection. Our approach fine-tunes MARBERTv2 on the Mawqif-XT dataset, exploring seven configurations including class weighting, regularization, data augmentation via back-translation, resampling strategies, and zero-shot inference with Qwen2.5-7B. Our best model by the primary metric Favg2 is M1 (Baseline), achieving 0.837 on the development set and 0.6704 on the official blind test set. Error analysis reveals that Favor→Against misclassification accounts for 84.6% of errors, driven by negative sentiment words co-occurring with supportive stances. We find that metric choice (Favg2 vs. Favg3) significantly impacts model selection decisions.

1 Introduction

Stance detection is the task of determining whether a text expresses Favor, Against, or None toward a given target. In Arabic social media, this task is particularly challenging due to dialectal variation, implicit sentiment, and sarcasm.

We participate in StanceEval-2026 (Albalawi et al., 2026a), the second Arabic stance detection shared task, using the Mawqif-XT dataset (Albalawi et al., 2026b). We focus on Track 1 (seen targets: COVID-19 Vaccine, Digital Transformation, Women Empowerment).

Our main contributions are: (1) a systematic comparison of seven fine-tuning strategies for MARBERTv2, (2) an empirical demonstration that back-translation augmentation degrades performance on dialectal Arabic, (3) a finding that the choice between Favg2 and Favg3 leads to different optimal model selection, and (4) a zero-shot evaluation of Qwen2.5-7B as a reference point.

Code: https://github.com/Elham-Alshankity/StanceEval-2026

2 Background

2.1 Task Setup

Given an Arabic tweet and a target topic, systems classify the expressed stance as one of three labels: Favor, Against, or None. The primary evaluation metric is Favg2—macro-average F1 over Favor and Against only—following the evaluation protocol introduced in prior stance detection work (Mohammad et al., 2016) and as defined by the StanceEval-2026 organizers (Albalawi et al., 2026a), with StanceEval-2024 providing relevant shared-task context (Alturayeif et al., 2024). Favg3 extends this to include the None class.

2.2 Dataset — Mawqif-XT

Mawqif-XT (Albalawi et al., 2026b) is collected from X.com and annotated for stance, sentiment, and sarcasm. The training set contains 3,502 tweets across three targets, with significant class imbalance: Favor (61.4%), Against (29.2%), None (9.5%). The development set contains 619 tweets.

2.3 Related Work

Researchers introduced Mawqif, the first Arabic dataset for target-specific stance detection, consisting of 4,121 tweets annotated for stance, sentiment, and sarcasm across three topics: the COVID-19 vaccine, women empowerment, and digital transformation. Their best model achieved an F1-score of 78%, and this is the same dataset family used in StanceEval, on which we apply our models (Alturayeif et al., 2023).

StanceEval-2024 provided the first Arabic stance detection shared task and reported 28 registered teams, 16 valid test submissions, and a highest classification F-score of 84.38% (Alturayeif et al., 2024).

MARBERTv2 is a pre-trained language model built on top of MARBERT, which was originally trained on a large collection of Arabic tweets in

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 613–616
October 28-29, 2026 ©2026 Association for Computational Linguistics

both Modern Standard Arabic and various dialects. It was further pre-trained on longer sequences to improve performance, making it well-suited to the nature of social media text compared to traditional BERT-based models (Abdul-Mageed et al., 2021).

The survey "Stance Detection on Social Media: State of the Art and Trends" reviews stance detection across NLP, web science, and social computing, including task definitions, target types, features, machine learning approaches, applications, and research gaps (ALDayel and Magdy, 2021).

3 System Overview

All encoder-based models fine-tune MARBERTv2 (Abdul-Mageed et al., 2021), a BERT-based model pre-trained on 1B+ Arabic tweets. We frame the task as sequence classification with input format: [TARGET] [SEP] [TWEET], enabling the model to condition its prediction on the specific target topic.

3.1 M1 — Baseline

Standard fine-tuning with cross-entropy loss, 5 epochs, batch size 32, learning rate 2e-5, no additional regularization.

3.2 M2 — Class Weights

Inverse-frequency class weights added to the loss function to address imbalance: Against (1.14), Favor (0.54), None (3.51). Uses WeightedTrainer with CrossEntropyLoss.

3.3 M3 — Dropout + LR Scheduling

Adds hidden dropout (0.2) and attention dropout (0.2) to M2 setup. Uses cosine LR scheduler with early stopping (patience=3), trained up to 10 epochs.

3.4 M4 — Back-Translation Augmentation

Generates 333 synthetic None-class samples via Arabic→English→Arabic back-translation using Helsinki-NLP/opus-mt models, doubling the None class from 333 to 666. Combined with M3 training setup.

3.5 M5 — Oversampling

Duplicates all 333 None-class samples via pd.concat, increasing None to 666 while keeping Favor (2,148) and Against (1,021) unchanged. Trains with balanced class weights computed via sklearn's compute_class_weight('balanced').

3.6 M6 — Undersampling

Reduces Favor and Against to 1,000 samples each via random sampling (random_state=42), retaining all 333 None samples. Final distribution: Favor=1,000, Against=1,000, None=333. Trained with M3 setup.

3.7 M7 — Qwen2.5-7B Zero-Shot

Evaluates Qwen/Qwen2.5-7B-Instruct with 4-bit quantization (BitsAndBytesConfig, nf4, float16) in zero-shot mode. The prompt provides target and tweet with label definitions. Output is parsed by keyword matching for Against, Favor, and None. Although the parser also supports the string "neutral" by mapping it to the official None label, Qwen is explicitly instructed to output only Against, Favor, or None. On the 619 development examples, neither the neutral-to-None mapping nor the unmatched-output fallback was triggered (0/619 for both cases).

4 Experimental Setup

4.1 Preprocessing

Stance labels are normalized to lowercase. NaN values in the stance column are mapped to 'none' via fillna before label encoding. No tweet text normalization is applied to preserve dialectal features.

4.2 Hyperparameters

Shared settings across encoder models: batch size=32, max sequence length=128, learning rate=2e-5, optimizer=AdamW (the default), seed=42. M3/M5/M6 additionally use cosine scheduler with early stopping patience=3.

4.3 Tools & Environment

Python 3.12, PyTorch, HuggingFace Transformers, scikit-learn, Helsinki-NLP MarianMT (back-translation). Training on NVIDIA Tesla T4 (16GB VRAM) via Google Colab.

5 Results and Analysis

5.1 Development Set Results

Our best model by Favg2 is M1 (Baseline) at 0.837, while M3 (Dropout+LR) achieves the highest Favg3 at 0.741. This discrepancy arises because Favg2 excludes the None class:

Optimizing for None (via class weights and augmentation) slightly reduces Against and Favor F1, lowering Favg2 while improving Favg3.

Model	Favg2	Favg3	Against	Favor	None
M1	0.837	0.699	0.783	0.890	0.424
M2	0.818	0.714	0.766	0.869	0.508
M3	0.827	0.741	0.772	0.881	0.571
M4	0.802	0.705	0.759	0.844	0.511
M5	0.816	0.712	0.753	0.878	0.505
M6	0.829	0.734	0.778	0.880	0.545
M7	0.664	0.544	0.681	0.647	0.305

Table 1: Results on the development set. Favg2 is the primary ranking metric; Favg3 includes None.

Notably, Qwen2.5-7B in zero-shot mode (M7, Favg2=0.664) performs substantially below fine-tuned MARBERTv2, confirming that task-specific fine-tuning on in-domain Arabic Twitter data remains essential even for large language models. For M7, Qwen was explicitly prompted to output one of the three official stance labels: Against, Favor, or None. The parser included a safeguard that maps "neutral" to None because the task defines the third class as None rather than Neutral. However, this safeguard was never triggered on the development set: no Neutral outputs and no unmatched outputs were observed across all 619 examples (0/619 for both cases). Thus, the reported M7 performance was not affected by either mapping or default behavior.

5.2 Official Test Results

We submitted M1 predictions on the blind test set, achieving an official Favg2 of 0.6704, ranking 20th on the final leaderboard. The gap between development (0.837) and test (0.670) performance suggests overfitting to development set characteristics during iterative model selection.

5.3 Analysis

Back-translation (M4) reduces Favg2 from 0.837 to 0.802. We attribute this to the dialectal nature of the tweets: Helsinki-NLP models trained primarily on Modern Standard Arabic produce paraphrases that alter the stance signal, introducing noise rather than useful variation.

Undersampling (M6) achieves the second-best Favg2 (0.829) with higher None recall than M1, suggesting that reducing majority class dominance is more effective than augmenting the minority class for this dataset.

5.4 Error Analysis

We analyze errors on the development set using M3. The dominant error type is Favor→Against misclassification, occurring 374 times (84.6% of 442 total errors). Three representative examples:

Pred \ True	Against	Favor	None
Against	–	374	–
Favor	–	–	–
None	59	–	–

Table 2: Partial confusion matrix showing dominant error patterns (M3, dev set).

1. الفساد الإداري أنهك البلاد ومازلنا نعتمد على السندات اليدوية لتحصيل الإيرادات، لابد من التحول الإلكتروني — True: Favor, Predicted: Against. The opening phrase carries strong negative sentiment, causing the model to predict Against despite the tweet ultimately supporting digital transformation.

2. بنك مصر الأفضل في مجال التحول الرقمي وخدمات الصراف الآلي — True: Favor, Predicted: Against. The model fails to recognize entity-level positive framing when the tweet praises a specific institution.

3. الحاجة الوحيدة اللي اتفدنا منها بكورونا --- التحول الحكومي الإلكتروني — True: Favor, Predicted: Against. The word كورونا activates negative associations, overriding the positive stance toward digital transformation.

These examples reveal that the model conflates sentiment polarity with stance, particularly when negative sentiment words co-occur with an ultimately positive stance toward the target.

6 Conclusion

We presented Nabra, a MARBERTv2-based system for Arabic stance detection in StanceEval-2026 Track 1. Through systematic comparison of seven configurations, we find that: (1) standard fine-tuning achieves the strongest Favg2, (2) back-translation augmentation hurts performance on dialectal Arabic, (3) metric choice between Favg2 and Favg3 leads to different model selection decisions, and (4) large zero-shot models remain behind task-specific fine-tuning. Error analysis identifies sentiment-stance conflation as the primary failure mode. Future work will address this through sentiment-aware training objectives and cross-target generalization.

Acknowledgments

We thank the StanceEval-2026 organizers for providing the Mawqif-XT dataset and evaluation in-

frastructure. Experiments were conducted using Google Colab.

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A Hyperparameter Details

Full training configuration for replication:

- All models: MARBERTv2 (UBC-NLP/MARBERTv2), num_labels=3, input=[TARGET][SEP][TWEET]
- M1: epochs=5, lr=2e-5, batch=32, no scheduler, no dropout override
- M2: same as M1 + WeightedTrainer (Against:1.14, Favor:0.54, None:3.51)
- M3: epochs=10, lr=2e-5, batch=32, cosine scheduler, dropout=0.2, early_stopping patience=3
- M4: M3 setup + 333 back-translated None samples (Helsinki-NLP opus-mt-ar-en + opus-mt-en-ar)
- M5: M3 setup + 333 duplicated None samples (oversampling)
- M6: M3 setup + undersampled Favor=1000, Against=1000, None=333
- M7: Qwen2.5-7B-Instruct, 4-bit nf4 quantization, max_new_tokens=5, temperature=0.0, zero-shot